Asem Ameen Qaid

Electrical Engineering Student · Power and Energy Systems · Embedded Systems and IoT

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Yogyakarta, Indonesia · Yemeni nationality

PROFILE

Third-year electrical engineering student at Universitas Islam Indonesia with a growing focus on power and energy systems. Builds and tests complete systems, from sensors and firmware to monitoring software, and documents clearly what has and has not been validated. President of UII Global, the university's international student organisation. Seeking international internships and research experience in energy systems, with the long-term goal of a Master's degree in Australia.

EDUCATION

B.Sc. Electrical Engineering, Universitas Islam Indonesia, Yogyakarta

Expected 2028

- CGPA 3.75 / 4.00
- Future Global Leaders Scholarship (FGLS), 2024 – present

ENGINEERING PROJECTS

Industrial Energy Efficiency Copilot

Oct 2026 – present

ANGEL Innovation Hackathon 2026 · Sustainable Energy track

- Lead Team PowerTech (four students, with a faculty supervisor) in building a multi-site industrial energy monitoring platform; responsible for system integration, the dashboard and the MQTT data pipeline.
- Software prototype with multi-site monitoring, baseline anomaly detection, energy-intensity (IKE) analysis, forecasting and a rule-based assistant; detected 3 of 3 inserted anomalies with no false alarms on a synthetic 30-day dataset.
- Acquisition unit (ESP32 with RS485 energy meter) designed and firmware written; bench validation planned for December 2026.

ESP32, PZEM meter, RS485, MQTT, Python dashboard

EE Studio: Interactive Virtual Engineering Laboratory

2026

Personal project · ee-studio-red.vercel.app

- Web platform of 37 interactive labs across 7 modules, including power systems (power flow, faults, protection) and renewable energy (PV, MPPT, wind, storage).
- Every model verified numerically before implementation; 485 automated backend tests; deployed on Vercel and Railway. Built with AI-assisted development under the author's direction and review.

React, TypeScript, FastAPI, NumPy, SciPy, pytest

Smart Electrical Panel Protection and Emergency Alert System

2026

Embedded Systems · final project

- Three-state protection system that disconnects the load within one 2-second cycle and sends SMS alerts; built on a soldered PCB and passed all six test scenarios.
- Resolved false flame alarms and GSM brown-outs by separating sensor and actuator power domains and buffering the SIM800L supply.

Arduino Nano, DHT11, MQ-2, IR flame sensor, SIM800L GSM, relay

IoT-Based Flood Early Warning System

2026

Disaster Mitigation course

- Breadboard prototype that classifies flood risk into three levels with local LED and buzzer alerts, and publishes readings over MQTT to a Node-RED dashboard.

ESP32, rain and water-level sensors, MQTT, Mosquitto, Node-RED

Speed Estimation Using the MPU6050 2026

Measurement and Instrumentation · final project, part 2 (pair project)

- Benchmarked two speed-estimation methods against 15 reference trials each: step counting achieved an RMSE of 0.161 m/s against 1.106 m/s for acceleration integration, explained by integration drift.

`MPU6050, Arduino Nano, complementary filter, Web Serial dashboard`

Dryer Control System Using the 8086 Microprocessor Jan 2026

Computer System and Microprocessor · final project

- Temperature control and over-pressure alarm in 8086 assembly; behaved as specified in all 15 simulated test cases.

`8086 assembly, 8255 PPI, ADC0804, LM35, MPX4115A, Proteus`

MQ135 Gas Sensor for CO_2 and Ethanol Measurement 2025

Measurement and Instrumentation · final project, part 1

- Calibrated MQ135 sensor and Python live interface; CO_2 survey of 15 campus locations from about 376 to 2,172 ppm.

`MQ135, Arduino Nano, Python`

LEADERSHIP

President, UII Global 2026 – present

- Lead the executive board and cross-functional teams that run cultural, educational and networking programmes for international students across the university's faculties.
- Coordinate programmes with the university administration, academic departments and external international partners.

Team Leader, Team PowerTech, ANGEL Innovation Hackathon 2026 Oct 2026 – present

- Run weekly progress reviews, a shared task board, Git-based peer review and a risk register for a multidisciplinary team.

Marketing Team Member, UII Global 2024 – 2026

- Supported event promotion, digital communication, social media and international student engagement.

TECHNICAL SKILLS

Programming	C, C++, Python, TypeScript, Intel 8086 assembly
Hardware	ESP32, Arduino, Intel 8086, 8255 PPI, sensor interfacing, GSM, RS485/Modbus
Software and tools	MATLAB, Simulink, Proteus, Quartus Prime, Arduino IDE, Node-RED, MQTT, FastAPI, React, Git
Methods	Sensor calibration, digital filtering, error analysis (RMSE), state-machine design, numerical verification
Research interests	Power systems, renewable energy integration, smart grid, energy management

HONOURS AND MEMBERSHIPS

- Future Global Leaders Scholarship, UII (2024 – present)
- First Medal, Faris Al-Ilqaa public speaking competition (high school)
- Founding Member, Jogja SWUG (SolidWorks User Group)
- Member, Google Developer Group (GDG)
- Student Representative, Almawirdy Board

LANGUAGES

Arabic	Native
English	Professional working proficiency
Indonesian	Intermediate

`Project reports and demonstrations: asemqaid.github.io/work`